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SPATTER MONITORING FOR ADDITIVE MANUFACTURING SYSTEMS

Abstract

An additive manufacturing system includes an energy delivery device configured to deliver energy to a build surface of a component to form a melt pool in the build surface of the component, a powder delivery device configured to direct a powder stream toward the melt pool, a spatter monitoring system, and a computing device configured to receive image data from the spatter monitoring system. The spatter monitoring system is configured to capture image data indicative of spatter, wherein spatter is material ejected from the melt pool. The computing device is configured to identify a spatter event based on the received image data and control at least one of the energy delivery device or the powder delivery device based on the determined spatter event.

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Background/Summary

TECHNICAL FIELD

[0001] The disclosure relates to additive manufacturing systems and techniques.

BACKGROUND

[0002] Additive manufacturing systems generate three-dimensional structures through addition of material layer-by-layer or volume-by-volume to form the structure, rather than removing material from an existing component to generate the three-dimensional structure. Additive manufacturing processes may be advantageous in many situations, such as rapid prototyping, forming components with complex three-dimensional structures, or the like. In some examples, additive manufacturing processes may utilize powdered materials, and may melt or sinter the powdered material together in predetermined shapes to form the three-dimensional structures. Variations in process conditions may result in defects and differences in microstructure in the three-dimensional structures.

SUMMARY

[0003] The disclosure describes additive manufacturing systems, and methods for operating additive manufacturing systems, that fabricate a component according to deposition parameters. An additive manufacturing system includes an energy delivery device that delivers energy to a build surface of a component to form a melt pool and a powder delivery device that directs a powder stream toward the melt pool. Spatter is material ejected from the melt pool, which may be liquid or at least partially melted. Spatter may be generated as a result of, for example, surface tension disruption of the melt pool and/or melt pool vaporization (e.g., pluming). Adverse effects may result from spatter, such as melted or partially melted material ejected from the melt pool adhering to the build surface and impacting the build quality of the layer, and/or spatter adhering to a delivery nozzle and interrupting the flow of the powder stream, or the like. Systems and techniques according to the present disclosure may include a spatter monitoring system and a computing device. The spatter monitoring system may capture image data indicative of spatter, and the computing device may identify spatter in the received image data and identify a spatter event based on the identified spatter. The computing device may control the powder delivery device and/or the energy delivery device based on the determined spatter event, such as by adjusting one or more parameters related to the powder delivery device (e.g., a carrier gas flow rate, a powder mass flow rate, or the like) and/or one or more parameters related to the energy delivery device (e.g., a focus or a power supplied to the energy delivery device, or the like). The computing device may further determine a relationship between the spatter event and either the deposition parameters that create the spatter event or the material parameters that result from the spatter event, such as by using machine learning techniques, that may be used to control the additive manufacturing system in the same or subsequent additive manufacturing processes. In this way, additive manufacturing systems described herein may control spatter to produce layers with improved build quality.

[0004] In some examples, the disclosure describes an additive manufacturing system which includes an energy delivery device configured to deliver energy to a build surface of a component to form a melt pool in the build surface of the component, a powder delivery device configured to direct a powder stream toward the melt pool, a spatter monitoring system configured to capture image data indicative of spatter, wherein spatter is material ejected from the melt pool, and a computing device. The computing device is configured to receive the image data from the spatter monitoring system, identify spatter in the received image data, identify a spatter event based on the identified spatter, and control at least one of the energy delivery device or the powder delivery device based on the determined spatter event.

[0005] In some examples, the disclosure describes a method for additive manufacturing that includes delivering, via an energy delivery device of an additive manufacturing system, energy to a build surface of a component to form a melt pool in the build surface of the component. The method includes delivering, via a powder delivery device of the additive manufacturing system, a powder stream toward the melt pool. The method further includes receiving, by a computing device, image data indicative of spatter from a spatter monitoring system of the additive manufacturing system, wherein spatter is material ejected from a melt pool formed on a component being manufactured by an additive manufacturing system. The method includes identifying, by the computing device, spatter in the received image data, and based on the identified spatter data, identifying a spatter event, and controlling, by the computing device and based on the determined spatter event, at least one of the powder delivery device or the energy delivery device.

[0006] The details of one or more examples are set forth in the accompanying drawings and the description below. Other features, objects, and advantages will be apparent from the description and drawings, and from the claims.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0007] FIG. 1A is a conceptual block diagram illustrating example spatter monitoring aspects of an example additive manufacturing system.

[0008] FIG. 1B is a conceptual block diagram illustrating example spatter monitoring image sensors of a spatter monitoring system and spatter analysis modules of a computing device of an additive manufacturing system.

[0009] FIG. 2 is a process flow diagram illustrating a mass flux and heat flux monitoring and control technique.

[0010] FIG. 3 is a flowchart illustrating an example method for fabricating a component using spatter monitoring.

[0011] FIG. 4A is a conceptual diagram illustrating an example machine learning model.

[0012] FIG. 4B is a conceptual diagram illustrating an example training process for a machine learning model.

[0013] FIG. 5A is a conceptual and schematic diagram illustrating a portion of an example an additive manufacturing system during an additive manufacturing technique that does not use spatter monitoring.

[0014] FIG. 5B is a conceptual and schematic side view diagram illustrating an example additively-manufactured component formed during the additive manufacturing technique of FIG. 5A.

[0015] FIG. 6A is a conceptual and schematic diagram illustrating a portion of the example additive manufacturing system during an additive manufacturing technique that employs spatter monitoring and control.

[0016] FIG. 6B is a conceptual and schematic side view diagram illustrating an example additively-manufactured component formed during the additive manufacturing technique of FIG. 6A.

DETAILED DESCRIPTION

[0017] The disclosure generally describes techniques and systems for monitoring spatter during a blown powder additive manufacturing technique, such as a directed energy deposition (DED) technique, and using an imaged spatter event to control fabrication of a component. During blown powder additive manufacturing, a component is built up by adding material in sequential layers, such that the final component is composed of a plurality of layers of material. In blown powder additive manufacturing techniques for forming components from metals or alloys, an energy source may direct energy at a substrate to form a melt pool, and a powder delivery device may deliver a powder to the melt pool. At least some of the powder at least partially melts and is joined to the

melt pool and, thus, the substrate.

[0018] Challenges may arise when performing additive manufacturing techniques with additive manufacturing systems. For example, spatter, which may be material ejected from the melt pool may adhere to undesired surfaces. Spatter may be generated in one or more spatter events, or continuously, as a result of one or more of a surface tension disruption (e.g., pressure imbalance between the melt pool and coverage gas), a vaporization of the melt pool (e.g., plumbing), or the like. The spatter event or events may indicate an unstable process that generally results in a lower quality build. For example, the spatter, which may be melted or partially melted powder, may adhere to one or more surfaces away from the melt pool, and may cause deleterious impacts. Spatter may adhere to the build surface of the component and cause a deposition anomaly, such as a defect in the topology of the build surface. Additionally, or alternatively, spatter may adhere to deposition equipment (e.g., a delivery nozzle) where it may interrupt delivery of the powder stream or the energy to the build surface, or may agglomerate and eventually fall off onto the build surface.

[0019] In accordance with aspects of this disclosure, an additive manufacturing system includes a spatter monitoring system. The spatter monitoring system includes one or more image sensors for capturing image data near the melt pool. The captured image data may be delivered to and received by a computing device. The computing device may identify, in the captured image data, a spatter event where material is ejected from the melt pool, and the computing device may adaptively control deposition based on the spatter event. For example, the computing device may, based on the determined spatter event, adjust one or more operational settings of the powder delivery device, the energy delivery device, or both.

[0020] In some examples, no spatter may be permitted, and any amount of material, however small, ejected from the melt pool may be identified as a spatter event. Alternatively, in some examples, a certain amount of spatter may be continuous, or unavoidable, or acceptable (e.g., determined to not interfere with the build). In such examples, the computing device may determine that the captured image data is indicative of spatter that exceeds a threshold (e.g., a non-zero threshold) for causing one or more deposition anomalies. Put differently, the computing device may determine that minor spatter does not constitute a spatter event, and only identify problematic amounts, distributions, or sizes of spatter (e.g., those that cause a deposition anomaly such as a topological feature on the build surface or buildup on a deposition head or delivery nozzle) to constitute a spatter event. Accordingly, the computing device may be configured to determine whether sensed spatter exceeds a threshold for causing deposition anomalies, and control the energy delivery device and/or the powder delivery device to reduce a magnitude or occurrence of the one or more deposition anomalies by reducing a magnitude or occurrence of the spatter. The computing device may use the determined spatter event to select or adjust deposition parameters to produce a layer having the desired parameters. In this way, additive manufacturing systems may more accurately produce a layer having the desired parameters compared to systems that so not use spatter monitoring to control deposition parameters.

[0021] The powder delivery device may be separate and discrete from each other, or may be combined as parts of a common deposition head. When parts of a common deposition head, the computing device may control the common deposition head to travel along a toolpath across the build surface to deposit a layer. In some examples, the computing device may also individually control the relative position of the powder stream to the melt pool by individually controlling the position of the powder delivery device and/or the energy delivery device within the common deposition head. For example, the computing device may determine that a spatter event has occurred or is occurring, and may adjust the position of the powder delivery device and/or the energy delivery device to reduce or eliminate spatter. In some cases, the deposition head may define a central longitudinal axis. The energy may be delivered coincident with or parallel to the central longitudinal axis.

[0022] In examples where the powder delivery device and the energy delivery device are parts of a common deposition head, the computing device may control the toolpath of the deposition head to account for spatter. For example, the computing device may adjust the toolpath to avoid (e.g., swerve around) a deposition anomaly. In some examples, the computing device may cause the deposition head to re-do (e.g., travel back over) a layer that contains spatter adhered to the layer with just the energy delivery device activated and without delivering powder to the melt pool. In this way, the computing device may cause the energy delivery device to scan a layer that contains spatter adhered to the build surface to re-melt and smooth a deposition anomaly caused by the spatter.

[0023] In some examples, the spatter monitoring system includes one or more image sensors configured to capture image data indicative of spatter. For example, the spatter monitoring system may include a melt pool monitor, which may be a thermal camera configured to observe the melt pool and capture infrared image data. The captured infrared image data captured by material ejected from the melt pool as spatter. In some examples, the spatter monitoring system may additionally, or alternatively, include one or more visual cameras configured to capture visual light (e.g., visible light) image data. Inclusion of multiple image sensors which capture different image data and/or are positioned to observe the melt pool at different distances or focuses with respect to the melt pool may be advantageous for capturing image data related to more types of spatter events or a more complete picture of a spatter event. Furthermore, captured image data by more than one type of camera (e.g., both infrared and visual spectra) may be advantageous to determination of more types of spatter events or a more complete picture of a spatter event.

[0024] As mentioned above, the additive manufacturing system may include a common deposition head, of which the energy delivery device are both components. The deposition head may define a central longitudinal axis. In some examples, one or more image sensors of the spatter monitoring system may be positioned on the central longitudinal axis, which may be referred to as an “on-axis” position. Positioning an image sensor on-axis may be advantageous for determining whether spatter will land outside of the surface of the melt pool, and thus cause a deposition anomaly.

[0025] However, it may be difficult to identify spatter using a spatter monitoring system which includes only an on-axis image sensor. For example, hot droplets of liquid positioned above the melt pool may not be identifiable by the computing device within the image data captured only by an on-axis image sensor. According to one or more examples of the present disclosure, spatter monitoring systems may include one or more image sensors that are not positioned on the central longitudinal axis of the deposition head, which may be called an “off-axis” position.

Advantageously, such an off-axis image sensor may, on its own or in combination with an on-axis image sensor, capture image data that is indicative of more types of spatter events or capture image data that more completely describes certain types of spatter events than spatter monitoring systems that use only an on-axis image sensor. For example, an off-axis image sensor may capture, and the computing device may identify and characterize, for example a spatter event that results in spatter building up on a delivery nozzle, which may not be recognized by a spatter monitoring system that includes only an on-axis image sensor.

[0026] The computing device or devices may be configured to control the powder delivery device and/or the energy delivery device according to a set of deposition parameters. For example, the computing device may set parameters controllable by the powder delivery device that include one or more of a carrier gas flow rate, a powder mass flow rate, and a delivery nozzle angle. Similarly, the computing device may set parameters controllable by the energy delivery device to deposit a layer. The parameters controllable by the energy delivery device may include one or more of a focus of the energy delivery device, a power supplied to the energy delivery device, and the like. Responsive to determining that a spatter event has occurred or is occurring, the computing device may adjust one or more parameters controllable by the powder delivery device and/or may adjust one or more parameters controllable by the energy delivery device. The adjustment(s) may reduce a

magnitude or frequency of occurrence of the spatter event or spatter events.

[0027] In one specific example, the deposition head may travel back and forth across the build surface laying down a series of adjacent tracks to deposit a layer on the build surface. As such, the deposition head may dwell at the ends of the track for a longer period of time than other portions of the track because the deposition head must slow down and change direction. At these portions of the track, the energy delivered may overheat and vaporize the melt pool, causing a spatter event. In some examples, the computing device may adjust the parameters of the energy delivery device by decreasing the power supplied to the melt pool to prevent overheating the melt pool and thus reduce the magnitude or occurrence of the spatter event. Other example causes of spatter events and control mechanisms to reduce their magnitude or occurrence are also considered.

[0028] Control (e.g., adjustment of parameters) of the powder delivery device and/or the energy delivery device in response to spatter may be performed in one or more ways by the computing device. In some examples, preset adjustments (e.g., slowing of the deposition head traveling down the toolpath, or reduction of the power supplied to the energy delivery device, or reduction of the mass flow rate of the powder, or the like) may be conducted, for example iteratively until the spatter event concludes. In some examples, the computing device may store a lookup table which stores both types or categories of spatter events and adjustments to settings of the powder energy delivery device and/or the energy delivery device. The computing device may perform an adjustment to the powder delivery device and/or the energy delivery device based on the recommended adjustment from the lookup table.

[0029] In one or more examples of the disclosure, the computing device may determine, via a machine learning model that takes the image data from the spatter monitoring system as input, one or more deposition parameters of the powder delivery device and/or the energy delivery device using one or more machine learning techniques. In this way, the computing device may be trained to execute adjustments to the settings of the powder delivery device and/or the energy delivery device based on previous spatter events and adjustments that were made to reduce or eliminate the frequency and/or magnitude of such spatter events.

[0030] Additive manufacturing systems and techniques of the present disclosure may be used to form any suitable component. For example, the additively-manufactured component may be a component of a gas turbine engine, or an additively-manufactured coating on a gas turbine engine component. Forming components or coatings of gas turbine engines in this way may reduce material waste, provide for increased quality, and/or impart desired temper, microstructure, or other material properties relative to gas turbine engine components formed by other manufacturing processes.

[0031] FIG. 1A is a conceptual block diagram illustrating an example additive manufacturing system **10** configured to monitor a gas flow during an additive manufacturing technique. In the example illustrated in FIG. 1A, additive manufacturing system **10** includes a computing device **12**, a powder delivery device **14**, an energy delivery device **16**, a powder flow monitoring system (PFMS) **18**, an optical system **54**, a stage **20**, a powder source **42**, a powder source mass sensor **44**, and a spatter monitoring system **48**.

[0032] Stage **20** is configured to position component **22** during an additive manufacturing process. In some examples, stage **20** is movable relative to energy delivery device **16** and/or energy delivery device **16** is movable relative to stage **20**. Similarly, stage **20** may be movable relative to powder delivery device **14** and/or powder delivery device **14** may be movable relative to stage **20**. Stage **20** may be configured to selectively position and restrain component **22** in place relative to stage **20** during manufacturing of component **22**.

[0033] Powder source **42** is the source of powder for powder stream **30**. Powder source **42** may include any suitable container or enclosure, such as a hopper, configured to hold powder. Powder source **42** also may include mechanism for entraining the powder in a gas flow. For instance, powder source **42** may be coupled to a gas source, which provides a gas flowing through powder

source **42** and entraining powder within the gas flow. Additionally, or alternatively, powder source **42** may include an agitator configured to agitate the powder and increase entrainment of the powder in the gas stream. System **10** may include a powder source mass sensor **44** associated with powder source **42**. Powder source mass sensor **44** may be configured to quantify loss of mass in the powder source **42** or, alternatively, a mass flow out of powder source **42**.

[0034] Powder source **42** is fluidically coupled to powder delivery device **14** via a flow path **46**. Flow path **46** may include any suitable structure(s) defining an enclosed flow between powder source **42** and powder delivery device, including conduit, pipe, tubes, or the like. Although not shown in FIG. **1A**, for at least part of flow path **46** between powder source **42** and nozzles of powder delivery device **14**, flow path **46** may split into multiple, parallel sections, e.g., one for each nozzle. Further, although not shown in FIG. **1A**, in some examples, flow path **46** may include one or more nozzles for controlling flow through flow path **46** as a whole or portions of flow path **46** (e.g., a section associated with a particular nozzle of powder delivery device **14**).

[0035] Powder delivery device **14** may be configured to deliver powder to selected locations of component **22** being formed via a powder stream **30**. Powder delivery device **14** may include one or more nozzles that each output powder and gas flow, such that the combined powder and gas flow defines powder stream **30** focused at a focus plane. The gas flow may include an inert gas, such as argon or nitrogen gas, that maintains an inert atmosphere near melt pool **32**. As powder delivery device **14** is movable in the z-axis shown in FIG. **1A** relative to component **22**, the focal plane of powder delivery device **14** also may be movable in the z-axis relative to component **22**, such that the focus plane may be controlled to be substantially coincident with build surface **28**.

[0036] In some examples, powder delivery device **14** may be mechanically coupled or attached to energy delivery device **16** to facilitate delivery of powder stream **30** and energy **34** for forming melt pool **32** to substantially the same location adjacent to component **22**. In some examples, powder delivery device **14** and energy delivery device **16** may be parts of a common deposition head. Energy delivery device **16** may include an energy source, such as a laser source, an electron beam source, plasma source, or another source of energy that may be absorbed by component **22** to form a melt pool **32** and/or be absorbed by powder in powder stream **30** to be added to component **22**. Example laser sources include a CO laser, a CO.sub.2 laser, a Nd:YAG laser, or the like. In some examples, the energy source may be selected to provide energy with a predetermined wavelength or wavelength spectrum that may be absorbed by component **22** and/or the powder to be added to component **22** during the additive manufacturing technique.

[0037] In some examples, energy delivery device **16** also includes an energy delivery head, which is operatively connected to the energy source. The energy delivery head may aim, focus, or direct energy **34** toward predetermined positions at or adjacent to a surface of component **22** during the additive manufacturing technique. As described above, in some examples, the energy delivery head may be movable in at least one dimension (e.g., translatable and/or rotatable) under control of computing device **12** to direct the energy toward a selected location at or adjacent to a surface of component **22**. In some examples, energy delivery device **16** may be translatable and/or rotatable in three dimensions.

[0038] As shown in FIG. **1A**, energy delivery device **16** may be arranged or configured such that energy **34** and powder stream **30** both exit from a common deposition head and are directed toward build surface **28**. For instance, energy **34** may pass through a central channel within the deposition head and exit a central aperture in the deposition head, while fluidized powder may flow through internal channels and powder delivery nozzle(s) for forming powder stream **30** and directing powder stream **30** toward build surface **28**. At least some of the powder in powder stream **30** may impact a melt pool **32** in component **22**, and at least some of the powder that impacts melt pool **32** may be joined to component **22**.

[0039] During additive manufacturing, component **22** is built up by adding material to component **22** in sequential layers. The final component is composed of a plurality of layers of material.

Energy delivery device **16** may direct energy **34** at first layer **24** to form melt pool **32**. Powder delivery device **14** may deliver powder stream **30** to melt pool **32**, where at least some of the powder at least partially melts and is joined to first layer **24**. Melt pool **32** cools as energy **34** is no longer delivered to that location of first layer **24** (e.g., due to energy delivery device **16** scanning energy **34** over the surface of first layer **24**). In this way the deposition head may traverse across build surface **28** (e.g., back and forth across build surface **28**) depositing tracks of material which form layer **26**.

[0040] System **10** may include both mass flow monitoring and heat flow monitoring. To provide mass flow monitoring, system **10** may include powder flow monitoring system (PFMS) **18**. PFMS **18** is configured to image at least a portion of powder stream **30** to detect powder flowing between powder delivery device **14** and build surface **28**. For example, PFMS **18** may include an illumination device and an imaging device. In some examples, the illumination device may include one or more light sources. For instance, the illumination device may include one or more structured light devices, such as one or more lasers. The illumination device is configured to illuminate a plane of powder stream **30** at image plane **38**, e.g., a plane substantially perpendicular to an axis extending between powder delivery device **14** and build surface **28**. The imaging device of PFMS **18** is configured to image at least some of the illuminated powder. The imaging device may have a relatively high data acquisition speed (e.g., frame rate), such greater than 1000 Hz.

[0041] To provide heat flow monitoring, system **10** may include one or more thermal image sensors configured to image at least a portion of melt pool **32** to detect parameters, such as size, temperature, or shape, of melt pool **32**. For example, a thermal sensor may be communicatively coupled to optical system **54** for observing thermal emissions around melt pool **32** and the same or a different thermal camera may monitor a size and/or temperature of melt pool **32**. Optical system **54** may include an imaging device and an associated optical train, which senses emissions at or near component **22** during the additive manufacturing technique. For example, optical system **54** may include a visible light imaging device, an infrared imaging device, or an imaging device that is configured (e.g., using a filter) to image a specific wavelength or wavelength range. The optical train may include one or more reflective, refractive, diffractive optical components configured to direct light to the imaging device. For example, the optical train may be configured to direct light from near component **22** and/or melt pool **32** to the imaging device.

[0042] In addition to mass flow monitoring and heat flow monitoring, system **10** includes spatter monitoring system **48**. Spatter monitoring system **48** may capture image data indicative of spatter. Spatter is material, such as solid powder, partially melted powder, or liquid material, that has been ejected from melt pool **32**. The image sensor or sensors of spatter monitoring system **48** may be indicative of spatter by being pointed at or above the surface of melt pool **32** to capture spatter when the spatter is airborne. Furthermore, the image sensor or sensors may be configured with a relatively fast shutter speed, such that the airborne spatter may be captured and analyzed. In some examples, the shutter speed may be 1/8000, or may be even faster to capture the spatter as frozen in a single place in the image data.

[0043] For example, spatter may be droplets of the melted layer **24** that form melt pool **32**. As used herein, ejected from melt pool **32** means material displaced from melt pool **32**, such as airborne material of material deposited discretely on other portions of build surface **28** or other parts of system **10**, such as powder delivery device **14** or energy delivery device **16**, where, among other problems, spatter may accumulate and eventually fall off onto build surface **28**.

[0044] System **10** may monitor spatter using spatter monitoring system **48** to reduce or avoid formation of deposition anomalies that may result from variation in effects, such as surface tension disruption of melt pool **32**, caused by parameters related to coverage gas flow, powder stream **30** and energy beam **34**. For example, the combination of settings of system **10** that include the mass flow rate of powder in powder stream **30**, the energy density and focus of energy beam **34**, the travel speed of powder delivery device **14** and/or energy delivery device **16** along a toolpath, and

the like, may result in melt pool vaporization, which may in turn cause plumbing where material is ejected due to boiling or frothing in the melt pool. Other causes of spatter in melt pool **32** are also considered. In any event, spatter indicates an unstable process that may result in one or more of delivery nozzle blockage, machine damage, component contamination, or the like.

[0045] To provide spatter monitoring, spatter monitoring system **48** may include melt pool (MP) monitor **15** and/or off-axis camera **21**, which may be the image sensors that capture image data indicative of spatter. In some examples, other numbers of image sensors may be used, for example a single image sensor, two image sensors, three image sensors, four image sensors, or even five image sensors. Spatter monitoring system **48** including multiple image sensors may be advantageous, because computing device **12** may be able to analyze and/or synthesize views of melt pool **32** at different angles and/or different zooms to identify and characterize spatter events, offering a more complete characterization and/or categorization of a spatter event.

[0046] In the illustrated example, MP monitor **15** is a thermal camera configured to capture infrared image data, and off-axis camera **21** is configured to capture visual light data. It may be advantageous to include both image sensors configured to capture visual light and infrared cameras configured to capture infrared image data. In this way, computing device **12** may be able to analyze and/or synthesize views of melt pool **32** from different modes to identify and characterize spatter events, offering a more complete characterization and/or categorization of a spatter event.

[0047] As mentioned above, energy delivery device **16** and powder delivery device **14** may be parts of a common deposition head that defines central longitudinal axis L. In the illustrated example, both MP monitor **15** and off-axis camera **21** are displaced from central longitudinal axis L in an “off-axis” configuration. In other examples, MP monitor **15** may be positioned on axis, where MP monitor **15** may be considered a part of both spatter monitoring system **48** and optical system **54**. Positioning at least one image sensor of spatter monitoring system **48** may advantageously allow for detection of spatter deposited on powder delivery device **14**, energy delivery device **16**, or another part of system **19**. As will be described further below, spatter event(s) captured by spatter monitoring system **48** may be correlated to anomalous or undesirable events and/or used to control powder delivery device **14** and/or energy delivery device **16** in a manner that reduces or avoids the anomalous or undesirable events.

[0048] Computing device model **12** includes machine learning (ML) model **67**. In some examples, ML model **67** may receive image data captured by MP monitor **15** and/or off-axis camera **21** as inputs to the machine learning model. In addition to data from MP monitor **15** and off-axis camera **21**, ML model **67** may receive data from optical system **54**, PFMS **18**, and/or other parameters (settings of powder delivery device **14**, settings of energy delivery device **16**, or other sensed data or settings of system **10**).

[0049] Computing device **12** is configured to control components and operation of system **10** and may include, for example, a desktop computer, a laptop computer, a workstation, a server, a mainframe, a cloud computing system, or the like. Computing device **12** may be communicatively coupled to, and configured to control, powder delivery device **14**, energy delivery device **16**, PFMS **18**, optical system **54**, stage **20**, powder source **42**, powder source mass sensor **44**, and/or spatter monitoring system **48** using respective communication connections. Although FIG. **1A** illustrates a single computing device **12** and attributes all control and processing functions to that single computing device **12**, in other examples, system **10** may include multiple computing devices **12**, e.g., a plurality of computing devices **12**.

[0050] Computing device **12** may be configured to control operation of powder delivery device **14**, energy delivery device **16**, adjustable z-stage **40**, and/or stage **20**, to position component **22** relative to powder delivery device **14**, energy delivery device **16**, PFMS **18**, and/or spatter monitoring system **48** during additive manufacturing processes. For example, as described above, computing device **12** may control stage **20** and powder delivery device **14**, energy delivery device **16**, and/or adjustable z-stage **40** to translate and/or rotate along at least one axis to position component **22**

relative to powder delivery device **14**, energy delivery device **16**, PFMS **18**, and/or spatter monitoring system **48**. Positioning component **22** relative to powder delivery device **14**, energy delivery device **16**, PFMS **18**, and/or spatter monitoring system **48** may include positioning a predetermined surface (e.g., build surface **28** to which material is to be added) of component **22** in a predetermined orientation relative to powder delivery device **14**, energy delivery device **16**, and/or spatter monitoring system **48**.

[0051] Computing device **12** may be configured to control system **10** to deposit layers **24** and **26** to form component **22** based on a set of deposition parameters. The set of deposition parameters may include energy, feed, and motion parameters that are configured to produce layers **24**, **26**, having various physical parameters, such as a height of layers **24**, **26**, and a density of layers **24**, **26**. For example, the set of deposition parameters may include power supplied, beam diameter (focus), beam profile, and wavelength of energy delivery device **16**; powder feed rate and gas feed rate of powder delivery device **14**; scan speed and deposition path of stage **20** relative to energy delivery device **16** and powder delivery device **14**; or any other operating parameters that may affect an amount and/or quality of material formed as layers **24**, **26**.

[0052] Computing device **12** may be configured to select deposition parameters that are configured to generate layers **24**, **26**, according to the desired physical parameters. As shown in FIG. **1A**, component **22** may include a first layer **24** and a second layer **26**, although many components may be formed of additional layers, such as tens of layers, hundreds of layers, thousands of layers, or the like. Component **22** in FIG. **1A** is simplified in geometry and the number of layers compared to many components formed using additive manufacturing techniques. Although techniques are described herein with respect to component **22** including first layer **24** and second layer **26**, the technique may be extended to components **22** with more complex geometry and any number of layers.

[0053] To form component **22**, computing device **12** may control powder delivery device **14** and energy delivery device **16** according to the set of operating parameters to form, on build surface **28** of first layer **24** of material, a second layer **26** of material. Computing device **12** may control energy delivery device **16** to deliver energy beam **34** to a volume at or near surface **28** to form melt pool **32**. For example, computing device **12** may control the relative position of energy delivery device **16** and stage **20** to direct energy to the volume. Computing device **12** also may control powder delivery device **14** to deliver powder stream **30** to melt pool **32**. For example, computing device **12** may control the relative position of powder delivery device **14** and stage **20** to direct powder stream **30** at or on to melt pool **32**.

[0054] Computing device **12** may control powder delivery device **14** and energy delivery device **16** to move energy **34** and powder stream **30** along build surface **28** in a pattern until layer **26** is complete. Computing device **12** may then control a z-axis position of stage **20** and/or powder delivery device **14** and energy delivery device **16** such that melt pool **32** will be formed on surface **36** of second layer **26**, and may control powder delivery device **14** and energy delivery device **16** to move energy **34** and powder stream **30** along build surface **28** in a pattern until layer **26** is complete.

[0055] As mentioned above, system **10** may be configured with various in-situ monitoring techniques, including mass flux monitoring, heat flux monitoring, and gas flow monitoring, to control an additive manufacturing process and a subsequent subtractive manufacturing process. FIG. **2** is a process flow diagram illustrating a mass flux and heat flux monitoring and control technique. The technique of FIG. **2** may be implemented by system **10** of FIG. **1A** and will be described with concurrent reference to FIG. **1A**.

[0056] Computing device **12** may be configured to control a powder feed rate output by powder source **42** (see top left of FIG. **2**). For instance, computing device **12** may be configured to control an agitator of powder source **42**, a gas flow rate of gas flowing through powder source **42**, a position of one or more valves within flow path **46**, or the like to control a powder feed rate output

by powder source **42**. Computing device **12** may be configured to receive data from one or more mass flow monitoring sensors, including PFMS **18** and/or powder source mass sensor **44**. Data received from powder source mass sensor **44** indicates a mass flow of powder from powder source **42** to powder delivery device. Data from PFMS **18** indicates a mass flow of powder in powder stream **30** between powder delivery device **14** to adjacent melt pool **32**.

[0057] Computing device **12** may calculate one or more mass flow-related metrics based on the data received from PFMS **18** and powder source mass sensor **44**. For example, one or more computing devices **12** may determine a capture efficiency by determining a fraction or percentage of powder from powder stream **30** that is captured by melt pool **32** and added to component **22**, e.g., by dividing the powder mass captured by melt pool **32** into the mass flow determined based on data received from PFMS **18**. Further, computing device **12** may determine an overall mass flux using the data received from PFMS **18** and/or powder source mass sensor **44**. One or more computing devices **12** then may use the overall mass flux as an input to the control algorithm used to control the powder feed rate output by powder source **42** (see top left of FIG. 2).

[0058] Similarly, computing device may be configured to control energy delivery device **16** to deliver energy beam **34** to first layer **24** to establish a given heat input (see bottom left of FIG. 2). For example, one or more computing device **12** may control one or more operating parameters of energy delivery device **16**, such as intensity, pulse rate, pulse width, or the like; one or more positional parameters related to energy delivery device **16**, such as dwell time at a location, a movement rate relative to first layer **24**, an overlap between adjacent passes of energy **34** across first layer **24**, a pause time between adjacent passes of energy **34** across first layer **24**, or the like to control heat input to system **10** (e.g., to melt pool **32** and component **22**).

[0059] Computing device **12** may be configured to receive data from one or more heat sensors, such as optical system **54** and/or MP monitor **15** of spatter monitoring system **48**. Computing device **12** may determine a cooling rate and associated heat from using data from optical system **54** and may determine a heat input into component using a size and/or temperature of melt pool **32** as observed by MP monitor **15**. Computing device **12** may be configured to determine an overall heat flux using these data. Computing device **12** may then use the overall heat flux as an input to the control algorithm used to control the energy delivery by energy delivery device **16** (see top left of FIG. 2).

[0060] Computing device **12** may be configured to monitor spatter through spatter monitoring system **48** to validate that system **10** is operating correctly. For example, spatter monitoring system **48** may capture image data using MP monitor **15** and off-axis camera **21**. Computing device **12** may identify spatter in the captured image data and determine that the identified spatter in the captured image data exceeds a threshold for causing one or more deposition anomalies. Based on the spatter exceeding the threshold for causing one or more deposition anomalies, computing device **12** may determine that the identified spatter constitutes a spatter event. Based on the determined spatter event computing device **12** may control one or both of powder delivery device **14** and energy delivery device **16**. For instance, computing device **12** may be configured to control a carrier gas flow rate, a powder mass flow rate, and/or a delivery nozzle angle of powder delivery device **14** and/or a focus, a power supplied, a pulse rate, a wavelength, or the like of energy beam **34** of energy delivery device **16** to end the spatter event. In this way, system **10** may include spatter monitoring by spatter monitoring system **48** to reduce or eliminate deposition anomalies and improve build quality of component **22**. In some examples, computing device **12** may control powder delivery device and/or energy delivery device **16** based at least partially on output(s) from ML model **67**, as will be further described below.

[0061] FIG. 1B is a conceptual block diagram illustrating example spatter monitoring sensors of spatter monitoring system **48** and spatter monitoring analysis and control modules of computing device **12** of additive manufacturing system **10**. Computing device **12** may be communicatively coupled to spatter monitoring system **48**, optical system **54**, and PFMS **18** to receive data from

spatter monitoring system **48**, optical system **54**, and PFMS **18**.

[0062] Spatter monitoring system **48** includes one or more image sensors configured to capture image data representative of a portion of system **10**. In some examples, the one or more image sensors may generate sensor data representative of spatter in the imaged portion of system **10**, and send the sensor data to computing device **12**. In the example of FIG. **1B**, spatter monitoring system **48** includes MP monitor **15** and off-axis camera **21**. MP monitor **15** is a thermal camera configured to capture infrared image data, and off-axis camera **21** is an image sensor configured to capture visual light data. However, spatter monitoring system **48** may include other image sensors that are configured to capture image data of other portions of system **10** (e.g., a different volume or a different focus) at the same or other modes. Computing device **12** may be configured to receive image data from the one or more image sensors of spatter monitoring system **48**.

[0063] In addition to data from spatter monitoring system **48**, computing device **12** may receive data indicative of various parameters related to powder flow, energy flow, gas flow, and the like of system **10**. For example, PFMS **18** may be configured to detect one or more parameters of system **10** related to powder flow, generate sensor data, and send the sensor data to computing device **12**. Sensor data from PFMS **18** may indicate a mass flow of powder in powder stream **30** between powder delivery device **14** to adjacent melt pool **32**. Similarly, optical system **54** may detect one or more parameters related to energy flow, generate sensor data, and send the sensor data to computing device **12**. Sensor data from optical system **54** may indicate a flow of energy in energy beam **34** between energy delivery device **16** and melt pool **32**.

[0064] Computing device **12** is configured to analyze the sensor data, including the image data from spatter monitoring system **48**, identify spatter in the image data, characterize and/or categorize a spatter event based on the image data, and control fabrication of a component based on the characterized gas flow. Computing device **12** includes a spatter monitoring module **64**, including a spatter event categorization module **65**, and an adaptive control module **66**, including a machine learning module **67**. FIG. **3** is a flowchart illustrating an example additive manufacturing method for fabricating a component using image data from spatter monitoring system **48**.

Operation of computing device **12** will be described with respect to the method of FIG. **3**. While computing device **12** will be described with respect to modules **64-67**, it will be understood that various modules may be performed by other computing devices. For example, spatter monitoring module **64** and adaptive control module **66** may be part of separate computing devices, such as specialized computing devices for implementing spatter monitoring and/or modelling and machine learning techniques.

[0065] The method includes controlling energy delivery device **16** and powder delivery device **14** (**80**). Adaptive control module **66** is configured to control energy delivery device **16** and powder delivery device **14** according to a set of deposition parameters which, as will be described further below, are at least partly based on data from spatter monitoring system **48**. For example, control module **66** may send control signals to energy delivery device **16** and powder delivery device **14** that control energy delivery device **16** and powder delivery device **14** to generate energy beam **34** and powder stream **30** according to particular parameters. As a result, energy delivery device **16** and powder delivery device **14** may deposit a layer on a build surface based on the control signals (**82**). For example, energy delivery device **16** may deliver energy to a build surface of a component to form a melt pool in the build surface of the component (**84**), and powder delivery device **14** may direct a powder stream toward the melt pool (**86**).

[0066] The method includes generating image data representative of spatter from spatter monitoring system **48** (**88**). For example, spatter monitoring system **48** may include MP monitor **15** that captures infrared image data and off-axis camera **21** that captures visual image data. MP monitor **15** and off-axis camera **21** may capture image data at particular periods to generate a temporal representation of spatter, such that the image data may approximate video data or other image data having a temporal component.

[0067] The method includes receiving image data indicative of spatter from spatter monitoring system **48 (90)**. In some examples, computing device **12** may receive the image data as powder delivery device **14** and energy delivery device **16** deliver powder stream **30** and energy beam **34**, respectively, to build surface **28**. For example, the image data may reflect local conditions of spatter in real time, such that computing device **12** may use the image data as immediate feedback. In some examples, computing device **12** may receive the sensor data after powder delivery device **14** and energy delivery device **16** deliver powder stream **30** and energy beam **34**, respectively, to build surface **28**. For example, the image data may reflect global conditions of spatter, such as for a particular layer or several layers, such that computing device **12** may use the image data as extended feedback for adjusting operation of powder delivery device **14** and/or energy delivery device **16** during a particular or subsequent build. In some examples, the method includes receiving additional sensor data, such as from at least one of optical system **54** or PFMS **18**.

[0068] The method includes determining a spatter event based on the received image data **(92)**. Spatter monitoring module **64** is configured to determine a spatter event based on the image data. For example, spatter monitoring module **64** may include spatter event categorization module **65**, which may store image data related to prior spatter events, and resulting quality defects or the like resulting from each of the stored spatter events. For examples, each previous spatter event stored in spatter event categorization module **65** may include a representation of a spatial, and optionally temporal, distribution of spatter near melt pool **32** of system **10** in the gas flow. Spatter monitoring module **64** may compare the received image data from spatter monitoring system **48** to the previous spatter events stored in spatter event categorization module, and may categorize the observe spatter as a particular type of spatter event. Spatter monitoring module **64** may generate a category for the spatter event, or may otherwise generate sufficient information from the image data for adaptive control module **66** to correlate characteristics of the spatter event with either/both deposition parameters that affect the spatter event or/and material parameters that are influenced by the spatter event.

[0069] In some examples, data associated with the spatter event may include parameters of powder delivery device **14** and energy delivery device **16**. For examples, parameters of powder delivery device **14** may include data related to carrier gas flow, such as a velocity, directionality, and/or composition of the gases in the carrier gas flow. Parameters of powder delivery device **14** may further include a powder mass flow rate, powder size/distribution, and material in powder stream **30**. Parameters of energy delivery device **16** may include a beam diameter of energy beam **16**, a power supplied to energy delivery device or the like. Parameters may be sensed or determined by analyzing various properties of the image data, alone or in combination with other sensor or model data. Additionally, or alternatively, the parameters may be settings of various components of system **10**. As will be described below, adaptive control module **66** may use one or more of these gas flow parameters as an input to a control algorithm used to control powder delivery device **14** and/or energy delivery device **16**.

[0070] In some examples, the image data received from spatter monitoring system **48** may include relatively unprocessed or minimally processed image data that may reflect spatter in real-time. For example, the spatter image data may be a visual display of the image data in real-time or near real-time that may visually indicate spatter generated in response to a change in one or more deposition parameters of powder delivery device **14** and/or energy delivery device **16**. In such examples, determining the spatter event based on the image data from spatter monitoring system **48** may include collecting and storing the image data in a form that can be optically analyzed for patterns at a set of deposition parameters or a change in patterns in response to a change in deposition parameters.

[0071] The method includes correlating the determined spatter event with deposition parameters that influence the gas flow and/or build quality **(94)**. Adaptive control module **66** may be configured to correlate the gas flow with the deposition parameters and/or material parameters

corresponding to build quality. For example, parameters of the carrier gas, powder flow, energy flow, or deposition head, such as the travel speed of the deposition head along a toolpath, may influence the spatter event (e.g., the amount, size, or distribution of spatter in the spatter event). [0072] In some examples, adaptive control module **66** may correlate the spatter event with one or more deposition parameters controllable by powder delivery device **14** and/or energy delivery device **16**. Adaptive control module **66** may be configured to analyze the spatter event with respect to deposition parameters and determine how a change in deposition parameters affects the spatter event. Particular spatter events may be associated with reduced build quality. For example, a turbulent gas flow at melt pool **32** may cause surface tension disruption at the surface of melt pool **32**, which may cause spatter and thereby reduce build quality. Adaptive control module **66** may be configured to identify a surface tension disruption as indicated by the image data and the carrier gas flow rate. The image data and the carrier gas flow rate data may be used to analyze an effect of changing deposition parameters of powder delivery device **14** and/or energy delivery device **16**.

[0073] In some examples, adaptive control module **66** may correlate the gas flow profile with one or more deposition anomalies. Deposition anomalies may include defects in the as-deposited layer **26** that may be indicated by the spatter event. For example, a particular spatter event may be differentiated and characterized, such that identification of the particular spatter event may indicate an increased likelihood of the deposition anomaly occurring. For example, a spatter event that includes spatter of a threshold size or concentration may meet a threshold for causing a topological defect on build surface **28** that will cause layer **24** and/or layer **26** to fall outside a specification window for thickness of the layer. In some examples, adaptive control module **66** is configured to determine whether the spatter event meets the threshold for corresponding to one or more deposition anomalies. Adaptive control module **66** is configured to control energy delivery device **16** and powder delivery device **14** to reduce a magnitude or occurrence of the one or more deposition anomalies. For example, as discussed above, adaptive control module **66** may be configured to determine how deposition parameters affect the spatter event. Such identified deposition parameters may be adjusted to produce a different spatter event or eliminate the spatter event altogether (e.g., no material added to melt pool **32** leaves melt pool **32**). In other words, adaptive control module **66** may adjust to the deposition parameters to produce a desired gas flow profile or avoid an undesired gas flow profile.

[0074] In some examples, adaptive control module **66** may be configured to correlate the spatter event with one or more structural features of additive manufacturing system **10** or component **22**. For example, particular structural features of additive manufacturing system **10**, such as clamps, tools, or walls of an enclosure, or component **22**, such as part geometry, may cause a spatter event. Adaptive control module **66** may be configured to identify changes in spatter caused by the structural features, such that any predicted spatter for similar structural features may be predicted and reduced or avoided.

[0075] In some examples, computing device **12** may be configured to use machine learning techniques to identify spatter, identify the identified spatter as correlating to a spatter event that deleteriously impacts build quality, correlate the identified spatter with one or more deposition parameters of system **10**, and/or identify an effect of structural features of system **10** or component **22** on spatter. Such machine learning techniques may enable computing device **12** to adapt and optimize deposition parameters using the observed gas flow indicated by the image data.

[0076] In some examples, machine learning model **67** may be configured to correlate the image data with spatter. In some examples, machine learning model **67** may correlate the identified spatter with the one or more deposition parameters using one or more machine learning techniques. Machine learning module **67** may be configured to determine one or more deposition parameters based on observed behavior indicated by the image data using model-based reinforcement learning. Machine learning module **67** may be configured to learn a model of spatter behavior based on the image data and use this model to make decisions about an effect of one or more deposition

parameters or change in one or more deposition parameters.

[0077] Machine learning module **67** may identify one or more deposition parameters related to powder delivery device **14** such as one or more of a carrier gas flow rate, a powder flow rate, or a delivery nozzle position, shape, or other parameter of powder delivery device **14**. Similarly, machine learning model **67** may identify one or more deposition parameters related to energy delivery device **16** such as a power or focus of energy delivery device **16**. Machine learning module **67** may further receive the image data from spatter monitoring system **48** and identify ways that a spatter event may be reduced (e.g., producing less spatter) or eliminated. For example, a spatter event that results in large droplets of spatter adhering to build surface may be associated with topological defects in the as-deposited layer. Machine learning module **67** may receive the spatter monitoring data from spatter monitoring system **48** for different sets of deposition parameters of energy delivery device **16** and powder delivery device **14**. As described above, the spatter monitoring data may include any combination of image data, parameters determined from the image data, model data that at least partially incorporates the image data, or settings of various parts of system **10**.

[0078] Machine learning model **67** may identify a machine learning algorithm that captures the relationship between the set of deposition parameters and the observed behavior as indicated by the gas flow profile. Models may include, but are not limited to, reinforcement learning models, such as deep reinforcement learning algorithms or traditional reinforcement learning algorithms. Machine learning model **67** may train the machine learning model using the gas flow profile. The model may learn the underlying patterns and relationships between the deposition parameters and the observed behavior as indicated by spatter. For example, machine learning model **67** may operate the model to improve a reward signal associated with the observed spatter behavior.

[0079] Machine learning model **67** may determine improved deposition parameters for a given set of conditions. For example, machine learning model **67** may use optimization methods, such as gradient-based methods or evolutionary algorithms, to tune the deposition parameters. Machine learning model **67** may further validate the model by analyzing spatter during fabrication of other layers and/or components.

[0080] Machine learning model **67** may be configured to correlate the spatter event with one or more deposition anomalies. Machine learning module **67** may be configured to determine one or more parameters of melt pool **32** and/or the as-deposited layer based on observed spatter behavior indicated by the image data using model-based reinforcement learning. Machine learning module **67** may be configured to learn a model of the spatter behavior based on the image data and use this model to make decisions about an effect of one or more material parameters or change in one or more material parameters.

[0081] Machine learning model **67** may identify one or more parameters that results from or are correlated with a spatter event. Machine learning model **67** may further receive sensor information that provides indication of such parameters, such as temperature data of melt pool **32**, topological data indicative of the surface of build surface **28**, a travel speed of a deposition head along a toolpath, or any other data that may provide an indication of deposition conditions affected by the spatter event. As described above, the data input into machine learning model **67** may include any combination of image data from spatter monitoring system **48**, parameters determined from the image data, model data that at least partially incorporates the image data, other sensed data from system **10**, or various settings of system **10**.

[0082] Machine learning model **67** may identify a machine learning algorithm that captures the relationship between the material parameters resulting from the spatter event. Models may include, but are not limited to, reinforcement learning models, such as deep reinforcement learning algorithms or traditional reinforcement learning algorithms. Machine learning model **67** may train the machine learning model using the spatter event and previous spatter events. The model may learn the underlying patterns and relationships between the material parameters and the observed

behavior as indicated by the spatter. For example, machine learning model **67** may operate the model to improve a reward signal associated with the observed spatter event.

[0083] Machine learning model **67** may determine improved deposition parameters for a given set of conditions. For example, machine learning model **67** may use optimization methods, such as gradient-based methods or evolutionary algorithms, to tune the deposition parameters. Machine learning model **67** may further validate the model by analyzing spatter events and spatter behavior during fabrication of other layers and/or components, and the resulting build quality that results from reduction or elimination of spatter events.

[0084] The method may include determining an adjusted set of deposition parameters based on the determined spatter event (**96**). Adaptive control module **66** may control powder delivery device **14** according to an adjusted set of deposition parameters that modify powder stream **30** based on the spatter event. Adaptive control module **66** may use correlations between the spatter event and the deposition parameters and correlations between the spatter event and the build quality to select a set of deposition parameters that modifies powder stream **30** and/or energy beam **34** to improve build quality and/or reduce or eliminate a spatter event. A spatter event may be considered reduced when a reduced number of spatter droplets are produced, or when the spatter droplets are reduced in size, or when a distance the spatter is ejected from melt pool **32** is reduced, or a combination of these. For example, adaptive control module **66** may generate control signals that cause powder delivery device **14** to modify any of a powder flow rate, a carrier gas flow rate, a delivery nozzle position or angle, or other parameter related to a velocity, directionality, or composition of the flow of powder stream **30** to achieve a powder stream that reduces or eliminates a spatter event. Similarly, adaptive control module **66** may generate control signals that cause energy delivery device **16** to modify any of a power, focus (beam diameter), working distance, pulse rate, dwell time, polarity, or other parameter related to a power, directionality, or wavelength of energy beam **34** to achieve an energy beam that reduces or eliminates a spatter event. In some examples, where powder delivery device **14** and energy delivery device **16** are parts of a common deposition head, the computing device may control the toolpath of the deposition head to account for spatter. For example, computing device **12** may adjust the toolpath to avoid (e.g., swerve around) a deposition anomaly. In some examples, computing device **12** may cause the deposition head to re-do (e.g., travel back over) a layer with only energy delivery device **16** activated and not powder delivery device **14** activated. In this way, energy delivery device **16** may deliver energy beam **34** to melt and smooth spatter adhered to a surface of component **22**.

[0085] As explained in FIG. **1B** with respect to machine learning model **67**, additive manufacturing systems described herein may be configured to use machine learning processes for determining a relationship between a spatter event and various upstream deposition parameters that cause the spatter event or downstream parameters of component **22** that result from the gas flow profile. FIG. **4A** is a conceptual diagram illustrating an example machine learning model according to one or more aspects of this disclosure. Machine learning model **67** may be an example of a deep learning model, or deep learning algorithm, trained to determine deposition parameters, material parameters, or structural features associated with or influenced by a spatter event. Computing device **12** and/or another device, may train, store, and/or utilize machine learning model **67**, but other devices of system **10** may apply inputs to machine learning model **67**. In some examples, other types of machine learning and deep learning models or algorithms may be utilized. For example, a convolutional neural network model of ResNet-18 may be used. Some non-limiting examples of models that may be used for transfer learning include AlexNet, VGGNet, GoogleNet, ResNet50, or DenseNet, etc. Some non-limiting examples of machine learning techniques include Support Vector Machines, K-Nearest Neighbor algorithm, and Multi-layer Perceptron.

[0086] As shown in the example of FIG. **4A**, machine learning model **67** may include three types of layers. These three types of layers include input layer **102**, hidden layers **104**, and output layer **106**. Output layer **106** includes the output from the transfer function **105** of output layer **106**. Input layer

102 represents each of the input values X1 through X4 provided to machine learning model **67**. In some examples, the input values may include any of the values input into the machine learning model, as described above. For example, the input values may include image data from spatter monitoring system **48**, various settings and parameters of energy delivery device **14**, various settings and parameters of energy delivery device **16**, various parameters related to component **22**, or another other data received by computing device **12**, as described above.

[0087] Each of the input values for each node in the input layer **102** is provided to each node of a first layer of hidden layers **104**. In the example of FIG. 4A, hidden layers **104** include two layers, one layer having four nodes and the other layer having three nodes, but fewer or greater number of nodes may be used in other examples. Each input from input layer **102** is multiplied by a weight and then summed at each node of hidden layers **104**. During training of machine learning model **67**, the weights for each input are adjusted to establish a relationship between image data from spatter monitoring system **48** and deposition parameters that generate spatter event represented by the image data. In some examples, one hidden layer may be incorporated into machine learning model **67**, or three or more hidden layers may be incorporated into machine learning model **67**, where each layer includes the same or different number of nodes.

[0088] The result of each node within hidden layers **104** is applied to the transfer function of output layer **106**. The transfer function may be linear or non-linear, depending on the number of layers within machine learning model **67**. Example non-linear transfer functions may be a sigmoid function or a rectifier function. The output **107** of the transfer function may be a set of deposition parameters, or a set of material parameters, that correspond to a gas flow profile depicted in the image data.

[0089] As shown in the example above, by applying machine learning model **67** to input data such as image data from spatter monitoring system **48**, processing circuitry of computing device **12** is able to determine a relationship between deposition parameters that produce, material parameters the result from, and/or structural parameter that affect the spatter event based on image data. Computing device **12** may utilize such features to determine a local coordinate system, to map the local coordinate system to a global coordinate system, and/or to generate representations of the spatter event.

[0090] FIG. 4B is a conceptual diagram illustrating an example training process for a machine learning model according to one or more aspects of this disclosure. Process **110** may be used to train machine learning model **67**. Machine learning model **67** may be implemented using any number of models for supervised and/or reinforcement learning, such as but not limited to, an artificial neural network, a decision tree, naïve Bayes network, support vector machine, or k-nearest neighbor model, CNN, RNN, LSTM, ensemble network, to name only a few examples.

[0091] In some examples, computing device **12** or another device trains machine learning model **67** based on a corpus of training data **112**. Training data **112** may include, for example, image data of previous spatter events, previous image data of spatter that did not meet a threshold for being determined a spatter event (e.g., did not result in a deposition anomaly), previous deposition parameters or material parameter data, and/or the like. Previous deposition parameter or material parameter data, for example, may include deposition parameters or material parameters associated with previous additive manufacturing processes. In some examples, the image data and/or deposition parameter or material parameter data may be generated by executing a natural language processing application on content of a plurality of operational records to automatically extract relevant nor desired training data. For example, by using an NLP model, computing device **12** may capture potentially confounding operation conditions or factors, which may be useful in determining a set of deposition parameters. Computing device **12** may use the NLP model to capture such details. In such an example, the machine learning model may analyze underlying characteristics associated with the powder, rather than powder flow factors, when determining a set of deposition parameters.

[0092] In some examples, training data **112** may include annotations identifying effects of deposition parameters indicated in image data of training data **112**. Training data **112** may include data from past additive deposition processes performed on components having different geometries, formed from different materials, formed under different conditions, and/or the like.

[0093] While training machine learning model **67**, computing device **12** may compare **119** a prediction or classification with a target output **117**. Computing device **12** may utilize an error signal from the comparison to train (learning/training **118**) machine learning model **67**. Computing device **12** may generate machine learning model weights or other modifications which computing device **12** may use to modify machine learning model **67**. For example, computing device **12** may modify the weights of machine learning model **67** based on the learning/training **118**.

[0094] FIG. 5A is a conceptual and schematic diagram illustrating a portion of an example additive manufacturing system **100** during an additive manufacturing technique that does not use spatter monitoring. Additive manufacturing system **100** of FIG. 5A may be an example of additive manufacturing system **10** of FIGS. 1A and 1B. Similar reference numerals indicate similar elements. Portions of additive manufacturing system **100** are not included in FIG. 5A. These portions will be described with respect to additive manufacturing system **10** of FIGS. 1A and 1B.

[0095] Additive manufacturing system **100** includes deposition head **156**, of which both energy delivery device **116** and powder delivery device **114** are components. In the illustrated example, energy delivery device **116** includes a laser mounted above or housed in a cavity within deposition head **156**. Energy delivery device **116** delivers energy beam **134** through an opening in deposition head **156** to form melt pool **132** on build surface **128** of previously deposited layer **124** of component **122**.

[0096] Powder delivery device **114** includes channels (not illustrated for clarity) formed within the wall of deposition head **156**. The channels are fluidically connected to delivery nozzles **158A**, **158B** (collectively “delivery nozzles **158**”) from which powder stream **130** is directed to melt pool **132**, adding material to melt pool **132** to form layer **126**. Layer **126** defines surface **136**.

[0097] Additive manufacturing system **100** includes MP monitor **115** and off-axis camera **121**, which may be portions of spatter monitoring system **48**. MP monitor **115** may be a thermal camera configured to image melt pool **132** and the surrounding area, and off-axis camera **121** may be configured to capture visual light data of a broader portion of system **100**, and may include within delivery nozzle **158B** within the field of view. MP monitor **115** and off-axis camera **121** may be configured to capture and send image data to computing device **12** (e.g., periodically capture and send image data).

[0098] System **100** of FIG. 5A is illustrated with spatter monitoring functionally turned off for illustration of the impacts of spatter when a spatter event occurs. During the spatter event of FIG. 5A, spatter particles **161A-161E** (collectively “spatter **161**”) are ejected from melt pool **132**. Spatter particle **161A** adheres to build surface **128A** at a location displaced from melt pool **132**. Spatter particle **161E** adheres to delivery nozzle **158B**, where it may interfere with the deliver of powder stream **130** to melt pool **132**, or may fall off at a later point in time and become adhered to build surface **128A**. Spatter particle **161C** adheres to surface **136** of as-deposited layer **126**.

[0099] FIG. 5B is a conceptual and schematic side view diagram illustrating example additively-manufactured component **122** formed during the additive manufacturing technique of FIG. 5A. As illustrated spatter particle **161A**, which adhered to build surface **128** during the process, may cool and harden to form topological defect **163A** on build surface **128**. Similarly, spatter particle **163C**, which formed on surface **136** of layer **126** during the process, may cool and harden to form topological defect **163C** on surface **136**. Topological defect **163A** may carry through as-deposited layer **126** to cause topological defect **163B**. Topological defects **163** may be deposition anomalies that cause layer **124**, layer **126**, and/or component **122** to fall outside a deposition window. As such, forming component **122** with additive manufacturing system **100** without spatter monitoring and control may result in a low-quality build.

[0100] FIG. 6A is a conceptual and schematic diagram illustrating a portion of the example additive manufacturing system **100** during an additive manufacturing technique that employs spatter monitoring and control. Additive manufacturing system **100**, when employing spatter monitoring and control as described herein, may form component **222** substantially free of defects related to spatter. For example, computing device **12** may adjust one or more parameters of system **100** to reduce or eliminate a spatter event to form component **222**. As illustrated, relative to FIG. 5A, computing device **12** may adjust the working distance (e.g., distance between deposition head **156** and build surface **228** (illustrated by the greater distance powder stream **130** must traverse to enter melt pool **232** in FIG. 6A), the shape and power of energy beam **134** of FIG (illustrated by the darker lines of energy beam **134** in FIG. 6A), and/or the angle of delivery nozzles **158** (more perpendicular to build surface **228** in FIG. 6A), or the like. As illustrated, the adjusted parameters may result in no spatter. Accordingly, component **222** of FIG. 6B, fabricated by the manufacturing process that includes spatter monitoring and control of FIG. 6A, may include layers **224** and **226** that are substantially free from deposition anomalies due to spatter.

[0101] The techniques described in this disclosure may be implemented, at least in part, in hardware, software, firmware, or any combination thereof. For example, various aspects of the described techniques may be implemented within one or more processors, including one or more microprocessors, digital signal processors (DSPs), application specific integrated circuits (ASICs), field programmable gate arrays (FPGAs), or any other equivalent integrated or discrete logic circuitry, as well as any combinations of such components. The term “processor” or “processing circuitry” may generally refer to any of the foregoing logic circuitry, alone or in combination with other logic circuitry, or any other equivalent circuitry. A control unit including hardware may also perform one or more of the techniques of this disclosure.

[0102] Such hardware, software, and firmware may be implemented within the same device or within separate devices to support the various techniques described in this disclosure. In addition, any of the described units, modules or components may be implemented together or separately as discrete but interoperable logic devices. Depiction of different features as modules or units is intended to highlight different functional aspects and does not necessarily imply that such modules or units must be realized by separate hardware, firmware, or software components. Rather, functionality associated with one or more modules or units may be performed by separate hardware, firmware, or software components, or integrated within common or separate hardware, firmware, or software components.

[0103] The techniques described in this disclosure may also be embodied or encoded in an article of manufacture including a computer-readable storage medium encoded with instructions. Instructions embedded or encoded in an article of manufacture including a computer-readable storage medium encoded, may cause one or more programmable processors, or other processors, to implement one or more of the techniques described herein, such as when instructions included or encoded in the computer-readable storage medium are executed by the one or more processors. Computer readable storage media may include random access memory (RAM), read only memory (ROM), programmable read only memory (PROM), erasable programmable read only memory (EPROM), electronically erasable programmable read only memory (EEPROM), flash memory, a hard disk, a compact disc ROM (CD-ROM), a floppy disk, a cassette, magnetic media, optical media, or other computer readable media. In some examples, an article of manufacture may include one or more computer-readable storage media.

[0104] In some examples, a computer-readable storage medium may include a non-transitory medium. The term “non-transitory” may indicate that the storage medium is not embodied in a carrier wave or a propagated signal. In certain examples, a non-transitory storage medium may store data that can, over time, change (e.g., in RAM or cache).

[0105] Various examples have been described. These and other examples are within the scope of the following examples and claims.

[0106] Example 1: An additive manufacturing system includes an energy delivery device configured to deliver energy to a build surface of a component to form a melt pool in the build surface of the component; a powder delivery device configured to direct a powder stream toward the melt pool; a spatter monitoring system configured to capture image data indicative of spatter, wherein spatter is material ejected from the melt pool; and a computing device configured to: receive the image data from the spatter monitoring system; identify, based on the received image data, a spatter event; and control at least one of the energy delivery device or the powder delivery device based on the identified spatter event.

[0107] Example 2: The additive manufacturing system of example 1, wherein, to identify the spatter event, the computing device is configured to: identify spatter in the received image data; and determine that the spatter identified in the received image data exceeds a threshold for causing one or more deposition anomalies to determine that the identified spatter constitutes the spatter event.

[0108] Example 3: The additive manufacturing system of any of examples 1 and 2, wherein the spatter monitoring system includes one or more off-axis image sensors.

[0109] Example 4: The additive manufacturing system of example 3, wherein the one or more off-axis image sensors include at least one thermal camera configured to capture infrared image data and one or more visual cameras configured to capture visual light data.

[0110] Example 5: The additive manufacturing system of any of examples 1 through 4, wherein the computing device is configured to control the powder delivery device according to a set of deposition parameters that includes one or more deposition parameters controllable by the powder delivery device, wherein the set of deposition parameters controllable by the powder delivery device include one or more of a carrier gas flow rate, a powder mass flow rate, and a delivery nozzle angle.

[0111] Example 6: The additive manufacturing system of example 5, wherein the computing device is further configured to: determine one or more deposition parameters controllable by the powder delivery device; and control, based on the one or more deposition parameters, the powder delivery device.

[0112] Example 7: The additive manufacturing system of any of examples 5 and 6, wherein the computing device is configured to determine, via a machine learning model that takes the image data from the spatter monitoring system as input, the one or more deposition parameters using one or more machine learning techniques.

[0113] Example 8: The additive manufacturing system of any of examples 1 through 7, wherein the computing device is configured to control the energy delivery device according to a set of deposition parameters that includes one or more deposition parameters controllable by the energy delivery device, wherein the set of deposition parameters controllable by the energy delivery device include one or more of a focus of the energy delivery device and a power supplied to the energy delivery device.

[0114] Example 9: The additive manufacturing system of example 8, wherein the computing device is further configured to: determine one or more deposition parameters controllable by the energy delivery device; and control, based on the one or more deposition parameters, the energy delivery device.

[0115] Example 10: The additive manufacturing system of any of examples 1 through 9, further comprising the component, wherein the component is a gas turbine engine component.

[0116] Example 11: A method for additive manufacturing includes delivering, via an energy delivery device of an additive manufacturing system, energy to a build surface of a component to form a melt pool in the build surface of the component; delivering, via a powder delivery device of the additive manufacturing system, a powder stream toward the melt pool, receiving, by a computing device, image data indicative of spatter from a spatter monitoring system of the additive manufacturing system, wherein spatter is material ejected from a melt pool formed on a component

being manufactured by an additive manufacturing system; identifying, in the received image data, spatter; identifying, by the computing device and based on the spatter in the received image data, a spatter event; and controlling, by the computing device and based on the identified spatter event, at least one of the powder delivery device or the energy delivery device.

[0117] Example 12: The method of example 11, further includes determining, via the computing device, that the spatter event corresponds to one or more deposition anomalies; and controlling, via the computing device, the energy delivery device and the powder delivery device to reduce a magnitude or occurrence of the one or more deposition anomalies.

[0118] Example 13: The method of any of examples 11 and 12, further comprising capturing, via one or more off-axis image sensors of the spatter monitoring system, image data indicative of spatter.

[0119] Example 14: The method of example 13, wherein capturing image data indicative of spatter with one or more off-axis image sensors includes capturing infrared image data with a thermal camera and capturing visual light image data with a visual light camera.

[0120] Example 15: The method of any of examples 11 through 14, further comprising controlling the powder delivery device according to a set of deposition parameters that includes one or more of a carrier gas flow rate, a powder mass flow rate, and a delivery nozzle angle.

[0121] Example 16: The method of example 15, further includes determining, via the computing device, one or more deposition parameters controllable by the powder delivery device; and controlling, via the computing device and based on the one or more deposition parameters, the powder delivery device.

[0122] Example 17: The method of any of examples 15 and 16, further comprising determining, via a machine learning model housed by the computing device that takes the image data from the spatter monitoring system as input, the one or more deposition parameters using one or more machine learning techniques.

[0123] Example 18: The method of any of examples 11 through 17, further comprising controlling the energy delivery device according to a set of deposition parameters that includes one or more of a focus of the energy delivery device and a power supplied to the energy delivery device.

[0124] Example 19: The method of example 18, further includes determining, via the computing device, one or more deposition parameters controllable by the energy delivery device; and controlling, via the computing device and based on the one or more deposition parameters, the energy delivery device.

[0125] Example 20: The method of any of examples 11 through 19, further comprising recording, via the computing device, the determined spatter event as part of a quality plan for the component.

Claims

1. An additive manufacturing system, comprising: an energy delivery device configured to deliver energy to a build surface of a component to form a melt pool in the build surface of the component; a powder delivery device configured to direct a powder stream toward the melt pool; a spatter monitoring system configured to capture image data indicative of spatter, wherein spatter is material ejected from the melt pool; and a computing device configured to: receive the image data from the spatter monitoring system; identify, based on the received image data, a spatter event; and control at least one of the energy delivery device or the powder delivery device based on the identified spatter event.

2. The additive manufacturing system of claim 1, wherein, to identify the spatter event, the computing device is configured to: identify spatter in the received image data; and determine that the spatter identified in the received image data exceeds a threshold for causing one or more deposition anomalies to determine that the identified spatter constitutes the spatter event.

3. The additive manufacturing system of claim 1, wherein the spatter monitoring system includes

one or more off-axis image sensors.

4. The additive manufacturing system of claim 3, wherein the one or more off-axis image sensors include at least one thermal camera configured to capture infrared image data and one or more visual cameras configured to capture visual light data.

5. The additive manufacturing system of claim 1, wherein the computing device is configured to control the powder delivery device according to a set of deposition parameters that includes one or more deposition parameters controllable by the powder delivery device, wherein the set of deposition parameters controllable by the powder delivery device include one or more of a carrier gas flow rate, a powder mass flow rate, and a delivery nozzle angle.

6. The additive manufacturing system of claim 5, wherein the computing device is further configured to: determine one or more deposition parameters controllable by the powder delivery device; and control, based on the one or more deposition parameters, the powder delivery device.

7. The additive manufacturing system of claim 5, wherein the computing device is configured to determine, via a machine learning model that takes the image data from the spatter monitoring system as input, the one or more deposition parameters using one or more machine learning techniques.

8. The additive manufacturing system of claim 1, wherein the computing device is configured to control the energy delivery device according to a set of deposition parameters that includes one or more deposition parameters controllable by the energy delivery device, wherein the set of deposition parameters controllable by the energy delivery device include one or more of a focus of the energy delivery device and a power supplied to the energy delivery device.

9. The additive manufacturing system of claim 8, wherein the computing device is further configured to: determine one or more deposition parameters controllable by the energy delivery device; and control, based on the one or more deposition parameters, the energy delivery device.

10. The additive manufacturing system of claim 1, further comprising the component, wherein the component is a gas turbine engine component.

11. A method for additive manufacturing, comprising: delivering, via an energy delivery device of an additive manufacturing system, energy to a build surface of a component to form a melt pool in the build surface of the component; delivering, via a powder delivery device of the additive manufacturing system, a powder stream toward the melt pool, receiving, by a computing device, image data indicative of spatter from a spatter monitoring system of the additive manufacturing system, wherein spatter is material ejected from a melt pool formed on a component being manufactured by an additive manufacturing system; identifying, in the received image data, spatter; identifying, by the computing device and based on the spatter in the received image data, a spatter event; and controlling, by the computing device and based on the identified spatter event, at least one of the powder delivery device or the energy delivery device.

12. The method of claim 11, further comprising: determining, via the computing device, that the spatter event corresponds to one or more deposition anomalies; and controlling, via the computing device, the energy delivery device and the powder delivery device to reduce a magnitude or occurrence of the one or more deposition anomalies.

13. The method of claim 11, further comprising capturing, via one or more off-axis image sensors of the spatter monitoring system, image data indicative of spatter.

14. The method of claim 13, wherein capturing image data indicative of spatter with one or more off-axis image sensors includes capturing infrared image data with a thermal camera and capturing visual light image data with a visual light camera.

15. The method of claim 11, further comprising controlling the powder delivery device according to a set of deposition parameters that includes one or more of a carrier gas flow rate, a powder mass flow rate, and a delivery nozzle angle.

16. The method of claim 15, further comprising: determining, via the computing device, one or more deposition parameters controllable by the powder delivery device; and controlling, via the

computing device and based on the one or more deposition parameters, the powder delivery device.

17. The method of claim 15, further comprising determining, via a machine learning model housed by the computing device that takes the image data from the spatter monitoring system as input, the one or more deposition parameters using one or more machine learning techniques.

18. The method of claim 11, further comprising controlling the energy delivery device according to a set of deposition parameters that includes one or more of a focus of the energy delivery device and a power supplied to the energy delivery device.

19. The method of claim 18, further comprising: determining, via the computing device, one or more deposition parameters controllable by the energy delivery device; and controlling, via the computing device and based on the one or more deposition parameters, the energy delivery device.

20. The method of claim 11, further comprising recording, via the computing device, the determined spatter event as part of a quality plan for the component.
